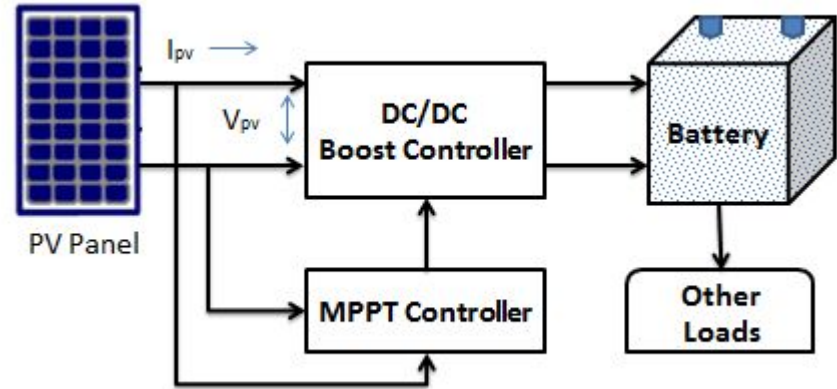


Circuito Maximum Power Point Tracking em FPGA para CubeSat

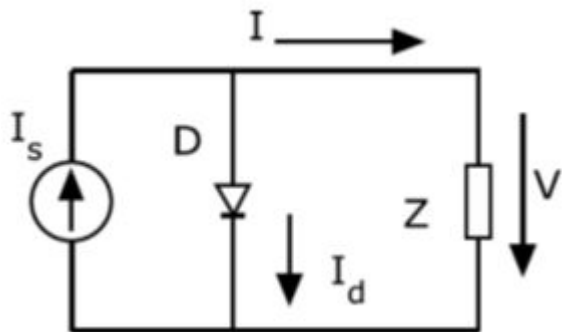
Marina Duailibe e Rebecca Quintino

Maximum Power Point Tracker (MPPT)



Características da célula

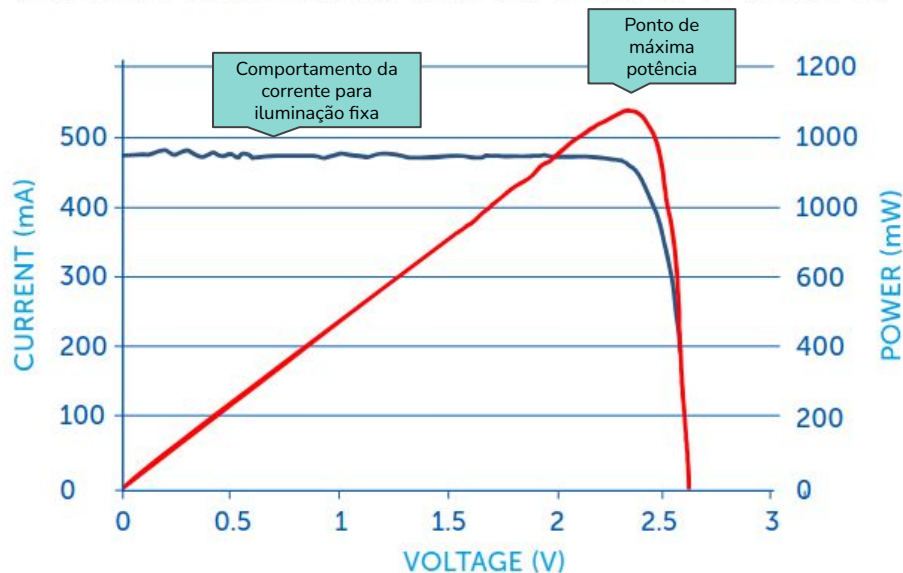
A corrente fornecida pela célula varia com a iluminação e temperatura.



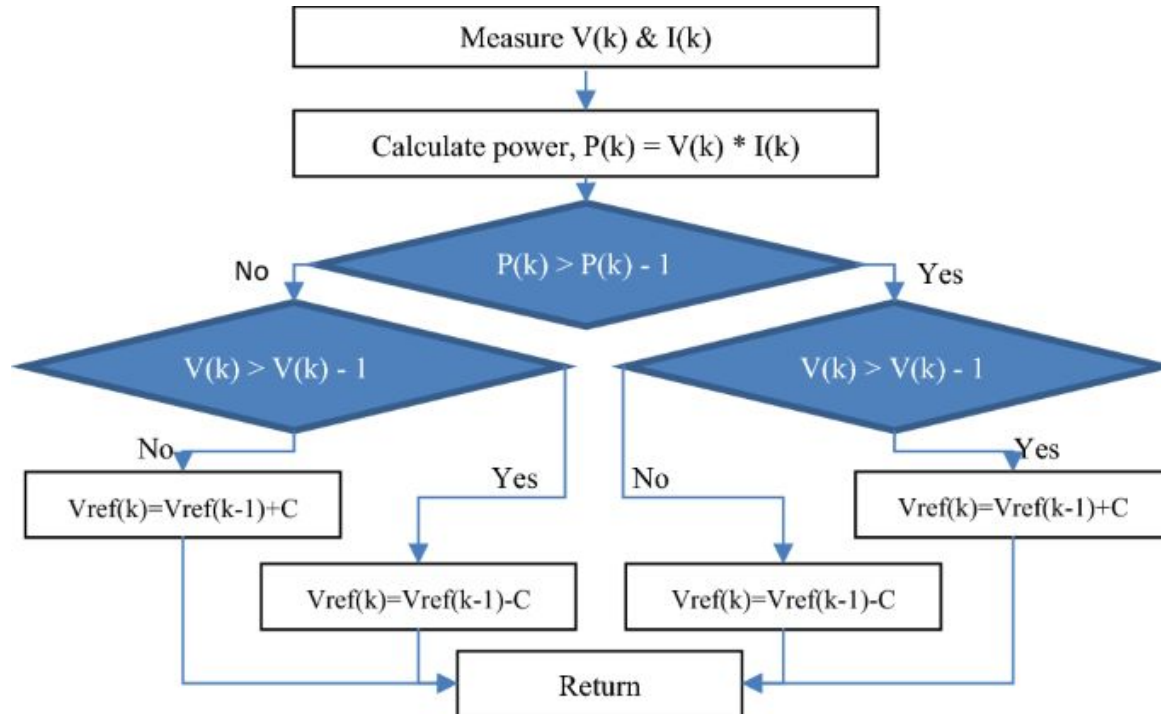
Circuito equivalente da célula

Typical Current-Voltage Characteristic

(CELL SIZE 26.5cm^2 , EFFICIENCY 29.5% @ AM0, 25°C)



Arquitetura Perturb and Observe

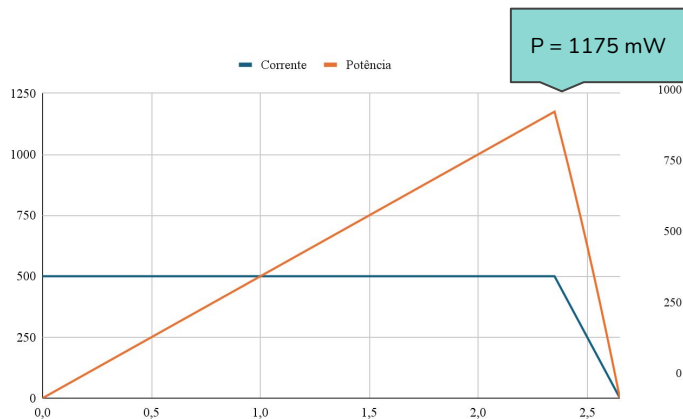




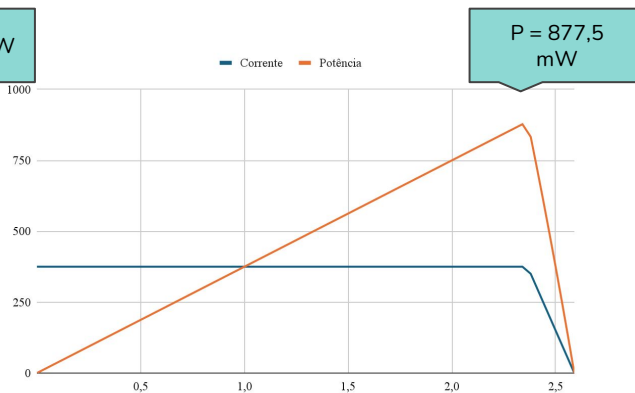
Validação dos cálculos teóricos

Comportamento para carga fixa (64 Ω) e iluminação variante

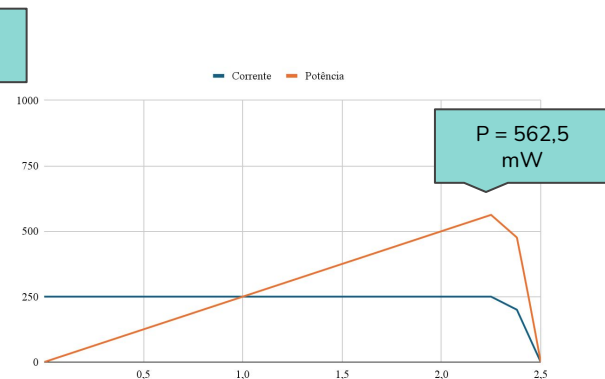
$$D = 1 - \sqrt{\frac{R_{in}}{R_{out}}}$$



Iluminação 100%			
D	Carga Rout	Rin	Vin boost(V)
0,7290064576	64	4,7	2,35



Iluminação 75%			
D	Carga Rout	Rin	Vin boost(V)
0,687750100	64	6,24	2,34



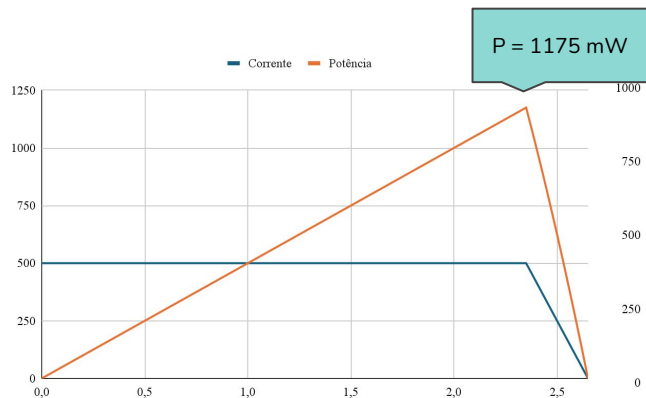
Iluminação 50%			
D	Carga Rout	Rin	Vin boost(V)
0,625	64	9	2,25



Validação dos cálculos teóricos

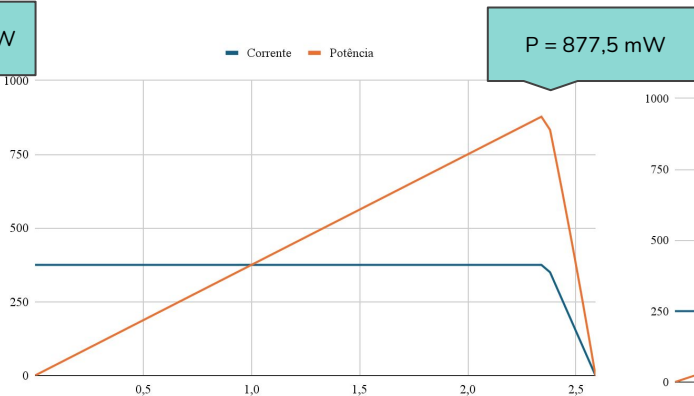
Comportamento para carga fixa (16 Ω) e iluminação variante

$$D = 1 - \sqrt{\frac{R_{in}}{R_{out}}}$$



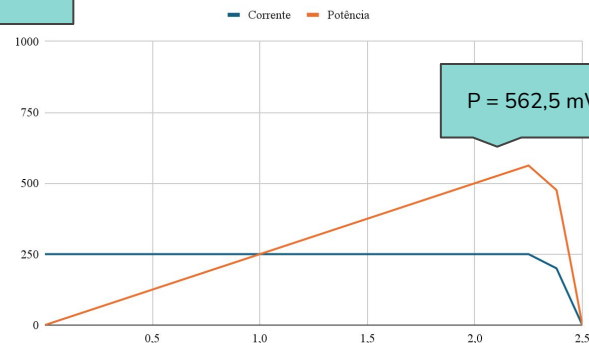
Iluminação 100%

D	Carga Rout	Rin	Vin boost(V)
0,4580129153	16	4,7	2,35



Iluminação 75%

D	Carga Rout	Rin	Vin boost(V)
0,375500200	16	6,24	2,34



Iluminação 50%

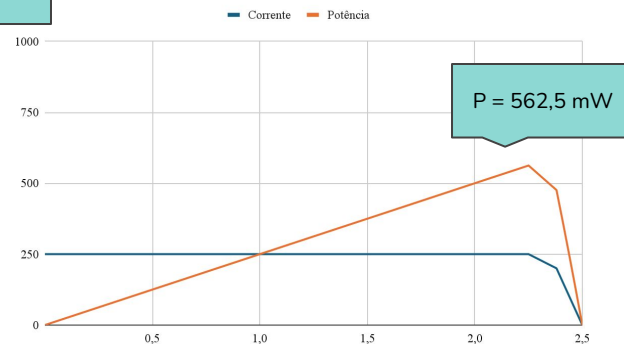
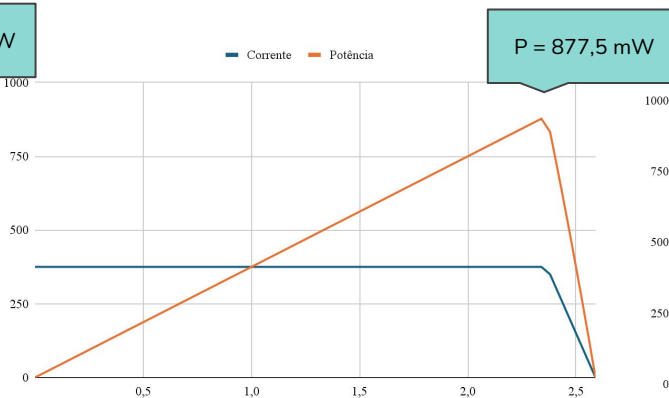
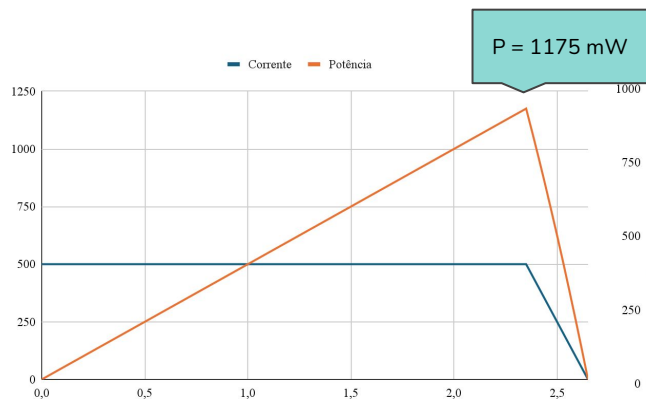
D	Carga Rout	Rin	Vin boost(V)
0,25	16	9	2,25



Validação dos cálculos teóricos

Comportamento para carga fixa (10 Ω) e iluminação variante

$$D = 1 - \sqrt{\frac{R_{in}}{R_{out}}}$$



Iluminação 100%

D	Carga Rout	Rin	Vin boost(V)
0,31443454	10	4,7	2,35

Iluminação 75%

D	Carga Rout	Rin	Vin boost(V)
0,210063293	10	6,24	2,34

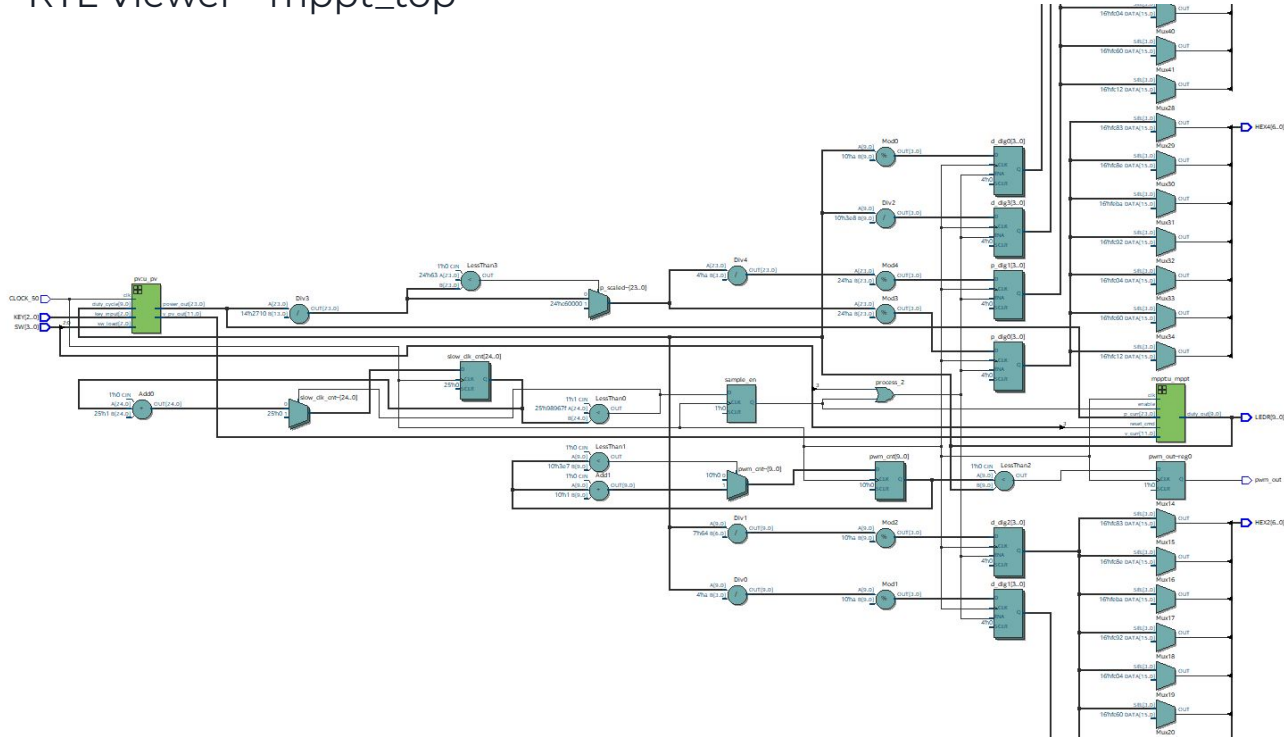
Iluminação 50%

D	Carga Rout	Rin	Vin boost(V)
0,0513167	10	9	2,25



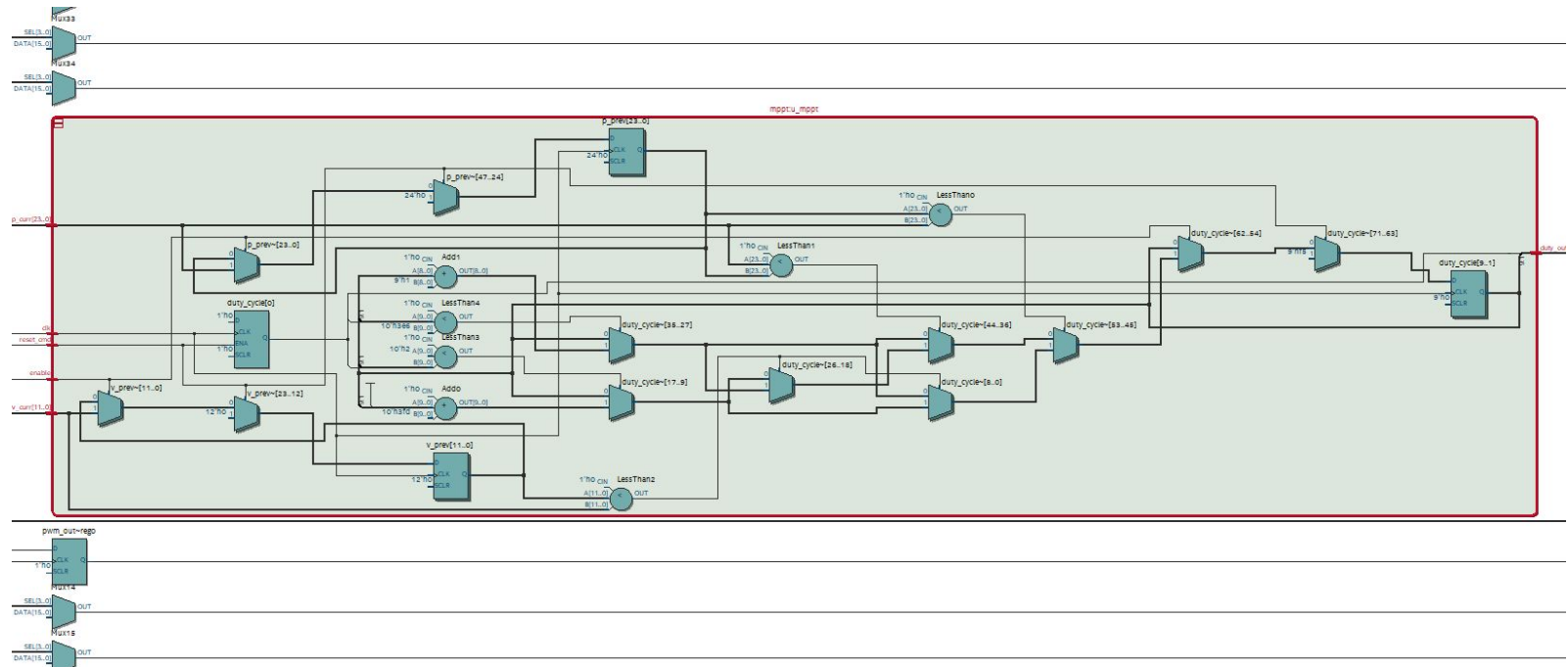
Implementação em VHDL

RTL Viewer - mppt_top



Implementação em VHDL

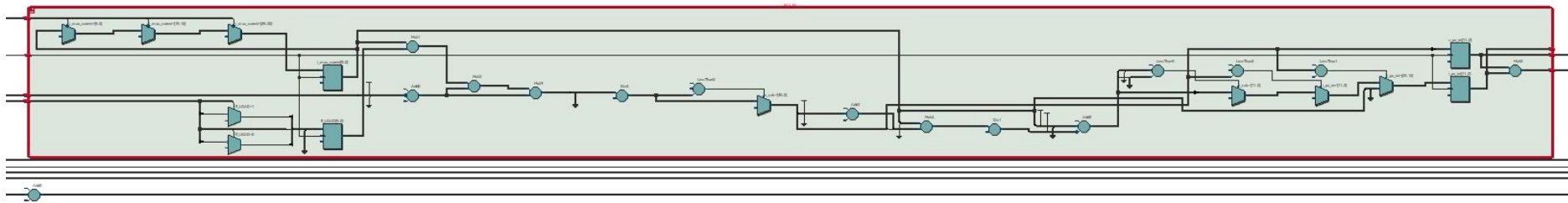
RTL Viewer - mppt





Implementação em VHDL

RTL Viewer - pv



Teste na FPGA

DevKit - DE1-SoC
FPGA - Cyclone V
Fabricante - Altera/Intel

Saídas

Display 7 segmentos:

HEX[0:2] = duty cycle

HEX[3:5] = potência atual

Entradas

Switches

SW0 - Recarga = 64 ohm

SW1 - Rcarga = 16 ohm

SW2 - Rcarga = 10 ohm

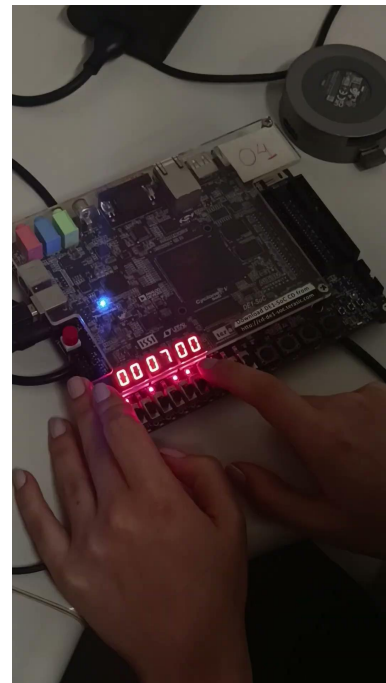
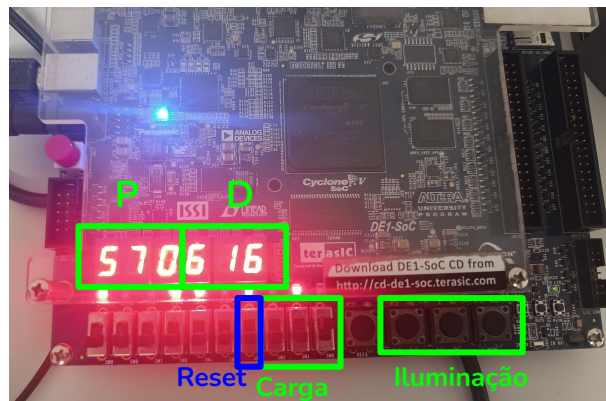
Botões

KEY0 - Iluminação 100% ($I_{pv} = 500 \text{ mA}$)

KEY1 - Iluminação 75% ($I_{pv} = 375 \text{ mA}$)

KEY2 - Iluminação 50% ($I_{pv} = 250 \text{ mA}$)

KEY3 - Reset





Comparação Teoria x Prática

Iluminação	Teoria						Prática					
	Carga 64 ohms		Carga 16 ohms		Carga 10 ohms		Carga 64 ohms		Carga 16 ohms		Carga 10 ohms	
	MPP (mW)	Duty Cycle	MPP (mW)	Duty Cycle	MPP (mW)	Duty Cycle	MPP (mW)	Duty Cycle	MPP (mW)	Duty Cycle	MPP (mW)	Duty Cycle
100%	1175	72,90%	1175	45,80%	1175	31,44%	990	73,00%	990	45,60%	990	31,60%
75%	877,5	68,78%	877,5	37,55%	877,5	21,01%	860	68,60%	870	37,20%	870	21%
50%	562,5	62,50%	562,5	25,00%	562,5	5,13%	570	63%	570	23,20%	580	14%

